

Reassessing EU Comparative Advantage: The Role of Technology*

Further Evidence from a Single Market Perspective

Filippo di Mauro[†]

Marco Matani[‡]

Gianmarco Ottaviano[§]

Abstract

Based on a sufficient statistics approach, we show how the state of technology of European industries relative to the rest of the world can be empirically assessed in a way that is simple in terms of computation, parsimonious in terms of data requirements, but still comprehensive in terms of information. The lack of systematic cross-industry correlation between export specialization and technological advantage suggests that standard measures of revealed comparative advantage only imperfectly capture a country's technological prowess due to the concurrent influences of factor prices, market size, markups, firm selection and market share reallocation. These findings offer policy insights relevant to the EU's external competitiveness debate, echoing several recommendations from the Draghi report. Achieving export specialization in key sectors requires more than just technological superiority, and this holds even more true for destinations outside the single market.

Keywords: *international trade, firm heterogeneity, monopolistic competition, multi-product firms, comparative advantage, European cross-country data, productivity*

JEL Classification: *B17, C19, C51, C80, D21, D43, F02, F12, F14, F61, L13, L25, O49*

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[†]CompNet. E-mail: filippodimauro1@me.com.

[‡]Aix-Marseille University, AMSE. E-mail: marco.matani@univ-amu.fr.

[§]Bocconi University, BAFFI, GREEN, IGIER, CEP, and CEPR. E-mail: gianmarco.ottaviano@unibocconi.it.

1 Introduction

The competitive position of the European Union has been a subject of extensive debate for decades. As highlighted by [Mengel and Ottaviano \(2024\)](#), the notion of ‘competitiveness’ has been central to numerous reports analyzing the economic performance of the European Union, with prominent recent contributions from [Monti \(2010\)](#), the [World Bank \(2012\)](#), the [World Economic Forum \(2012\)](#), the [European Investment Bank \(2016\)](#), the [European Commission \(2023\)](#), the [European Investment Bank \(2024\)](#), and the reports by [Draghi \(2024\)](#) and [Letta \(2024\)](#). These reports utilize the term ‘competitiveness’ with diverse interpretations, reflecting the inherent ambiguity of a concept that is both compelling and elusive ([Krugman, 1994](#)).

[Mengel and Ottaviano \(2024\)](#) identify two primary conceptualizations of competitiveness. The first, and more prevalent, definition pertains to ‘internal competitiveness’ – a country or region’s capacity for self-improvement, independent of direct comparison with rival economies. Within this framework, the authors define competitiveness in terms akin to productivity or efficiency, albeit with subtle variations in emphasis. These variations often arise from differing priorities. For the [World Bank \(2012\)](#), this form of competitiveness is largely viewed through the lens of GDP growth, whereas the World Economic Forum defines it more broadly as an economy’s ability to sustain ‘high and rising living standards.’ The reports by [European Investment Bank \(2016\)](#) and [European Investment Bank \(2024\)](#) associate it with wealth creation, and [Draghi \(2024\)](#), supported by [Letta \(2024\)](#), emphasizes the upscaling of innovative startups through enhanced integration. Beyond these definitional nuances, some reports argue that genuine competitiveness requires an economy to not only achieve economic efficiency targets but also to simultaneously address crucial objectives related to equality and sustainability.

The second definition of competitiveness, employed by these reports and the focus of our attention, is ‘external competitiveness’ – the capacity to effectively compete with other entities. This perspective gains prominence when the EU is considered within the context of the global economy. Here, the emphasis lies on the extent to which the Union enables its firms to maintain or expand their market share in international markets. The ultimate objectives associated with achieving external competitiveness also vary. Certain reports, such as that by the [World Bank \(2012\)](#), primarily outline the necessary actions for the EU to enable its firms to offer goods and services that are attractive to other nations. Consequently, while not explicitly advocating for a cooperative international environment, these reports maintain a non-zero-sum perspective. Conversely, other reports, including those by [Monti \(2010\)](#) and [European Commission \(2023\)](#), express greater concern regarding the penetration of foreign firms into the European market and the Union’s ability to preserve a dominant position in key ‘strategic’ industries. Similarly, [Draghi \(2024\)](#) underscores the challenge posed by the EU’s declining technological leadership compared to the United States and increasingly China in several sectors, advocating for proactive industrial policy to reverse this trend. In this regard, while assessing Europe’s relative technological prowess appears crucial for evaluating the costs and benefits of proposed industrial policy interventions, establishing a computationally simple, data-parsimonious, yet informationally comprehensive method for this assessment remains an unresolved issue.

In this paper we show how this hurdle can be overcome by applying the ‘sufficient statistics’ approach recently developed by [Huang and Ottaviano \(2024\)](#) in the case of China.¹ The idea is to leverage international trade data and firm information to infer a country’s relative technological prowess from its sectoral export specialization with respect to the rest of world. This is akin to Balassa’s idea of using export specialization to ‘reveal’ comparative advantage ([Balassa, 1965](#)). If with trade (‘ex post’) a country is relatively specialized in exporting in some sectors, then it must be that without trade (‘ex ante’) in those sectors it would be able to quote lower relative autarky prices than its trading partners.

Balassa’s insight is based on traditional trade models, in which product markets are perfectly competitive and production exhibits constant returns to scale so that across sectors relative prices coincide with relative costs. While export specialization reflects relative cost advantage, it cannot be used to identify relative technology and relative factor prices as separate sources of such advantage ([Deardorff, 1980](#)). Identification

¹See [Arkolakis et al. \(2012\)](#) and [Arkolakis et al. \(2019\)](#) for a sufficient statistics approach to welfare analysis in the wake of [Chetty \(2009\)](#).

faces additional challenges with scale economies and imperfectly competitive product markets. In this case, revealed comparative advantage may not coincide with relative cost advantage due to variable markups and scale effects as countries' relative market sizes disturb the transmission from relative costs to export specialization through relative prices. Firm heterogeneity further complicates the picture as endogenous trade-induced firm selection and market share reallocation add new disturbances, as highlighted for instance by [Bernard et al. \(2007\)](#) and [Melitz and Ottaviano \(2008\)](#).²

All this has implications for the interpretation of the Balassa index of 'revealed comparative advantage' (BRCA), defined as the ratio of a sector's share of exports for a given country to its share for the rest of the world. Values of a country's BRCA larger (smaller) than one highlight the sectors in which the country enjoys a relative price advantage (disadvantage), but it is silent about the role of the different possible sources of the price advantage. In particular, it cannot be used to assess the extent to which the relative price advantage is driven by relative technological prowess. In other words, a country's relative export specialization in a sector is a sufficient statistics for its relative price advantage, but not for its relative state of technology.³

[Huang and Ottaviano \(2024\)](#) push Balassa's logic one step beyond. They show how the structure of a two-country general equilibrium model, encompassing all the aforementioned disturbances, can be used to define a set of sufficient statistics that together make it possible to recover a country's state of technology relative to a reference country from trade data and firm information after filtering out the possible influences on exports of factor prices, market size, markups, selection and reallocation. Their model could be applied 'brute force' by using its equilibrium relations to calibrate the different countries' sectoral states of technology. Given the large number of equations and parameters involved (determined by the number of countries times the number of sectors), calibration requires targeting a long list of empirical moments as well as inverting a high-dimensional system of equations, which is clearly demanding in terms of both data and computation. What [Huang and Ottaviano \(2024\)](#) show is that a country's relative state of technology in a given sector can be alternatively assessed by relying on a single equilibrium equation and a very short list of empirical moments ('sufficient statistics'), mostly related to the country of interest, without having to calibrate the entire model.⁴ These moments are the country's relative market size and relative factor prices, together with its sector-specific export propensity (i.e. share of firms that export), export intensity (i.e. export over total revenues), trade freeness (i.e. international market accessibility), and heterogeneity in firm productivity.

In particular, the sufficient statistics approach proposed by [Huang and Ottaviano \(2024\)](#) offers the opportunity to compute a handy index of a country's (revealed) 'relative state of technology' (RST) with respect to the rest of the world. This is larger than one in sectors where the country has a better state of technology than the rest of the world, which is defined as the minimum productivity firms can expect to achieve when entering the market ([Eaton and Kortum, 2002](#)) and can be considered as an inverse measure of the country's distance from the global technological frontier. Their sufficient statistics approach also allows disentangling how factor prices, market size, markups, selection and reallocation affect a country's relative productivity across sectors as the country opens up to trade.

Leveraging [Huang and Ottaviano \(2024\)](#), we hereby provide the first comprehensive evaluation of Europe's RST in manufacturing with special emphasis on its two biggest western economies (France and Germany) and its biggest eastern economy (Poland), also supplemented by specific comparisons with the United States and China. Apart from being interesting in itself given the public debate, Europe provides an ideal setting to implement the sufficient statistics analysis thanks to the unique richness and quality of available harmonized micro-aggregated moments for several sectors in several countries. In particular, focusing on Europe allows us to leverage CompNet, a micro-aggregated dataset of productivity indicators and their drivers, with special emphasis on trade. The final aim is twofold. First, by computing the RST, we want to uncover where Europe's relative technological advantage currently lies. Second, by comparing the patterns of RST and BRCA, we

²Other disturbances may arise from policies targeting export activity in specific industries, institutional features, positioning within global value chains, geographical or spatial characteristics. We try to control for these aspects when comparing our measure of the relative state of technology with alternative measures of relative productivity (see Tables 1 and 2), but we leave a deeper investigation of their potential role to future research.

³See [Costinot et al. \(2012\)](#), [French \(2017\)](#) and, in a framework more akin to ours, [Del Gatto \(2018\)](#) for a related discussion.

⁴See, e.g., [Costinot et al. \(2012\)](#), [Lagakos and Waugh \(2013\)](#), [Levchenko and Zhang \(2016\)](#), [Costinot et al. \(2016\)](#), [French \(2017\)](#), [Del Gatto \(2018\)](#) and [Redding and Weinstein \(2018\)](#) for alternative approaches.

want to unveil the relative importance of factor prices, market size, market power, selection and reallocation in misaligning Europe’s relative export specialization and its relative technological advantage.

We find that Europe has better state of technology than the rest of the world ($RST > 1$) in all manufacturing industries, but especially in wearing apparel, food, basic metals and furniture, and non-metallic mineral products. Among these industries, however, only in food and non-metallic mineral products Europe also exhibits export specialization ($BRCA > 1$). In other sectors, such as motor vehicles, wood, paper products, and machinery, Europe has strong export specialization despite less pronounced technological advantage. By contrast, the EU holds a BRCA with respect to the US in all industries. When compared with China instead, the EU remains technologically more advanced across all sectors, but its BRCA is consistently lower than with respect to the RoW.

In general, the lack of systematic correlation between RST and BRCA confirms that revealed comparative advantage is an imperfect measure of a country’s technological prowess due to the concurrent influences of factor prices, market size, markups, firm selection and market share reallocation. In particular, with respect to misalignment, we find that, after controlling for technological prowess, higher unit input price and more heterogeneity as well as lower trade freeness and smaller relative market size reduce a country-industry’s relative export performance. We also observe that the role of these elements driving the discrepancy is more salient when exporting to global destinations than within the single market, and that in general firm heterogeneity became more prominent over time.

The implications of these results for policy formulation are pertinent to the ongoing discussion regarding the external competitiveness of the EU, aligning with several recommendations outlined in the Draghi report. The pursuit of export specialization in strategically important industries demands a multifaceted approach that extends beyond the acquisition of a technological advantage. Crucially, it also involves establishing appropriate conditions to effectively leverage this advantage, encompassing affordable access to necessary production factors (and intermediate inputs), the development of a significant domestic market for related outputs, and the promotion of vibrant industry dynamics conducive to the introduction of such products. Consequently, policy attention should be directed towards the development of workers’ skills, the mobilization of investment capital, the clever design of supply chains, the further deepening of single market integration (also fostering labor and capital mobility), and the facilitation of firm entry and exit, as well as the provision of adequate incentives for the most efficient firms that are more likely to succeed in international markets.

The rest of the paper is organized in four sections. Section 2 describes the data we use. Section 3 presents the theoretical framework of the analysis. Section 4 reports the empirical results. Section 5 concludes.

2 Data

The analysis exploits two main data sources. The first is the 9th Vintage of the CompNet dataset, an unbalanced panel dataset covering non-financial corporations across 22 European countries (CompNet, 2023). This dataset is gathered by the Competitiveness Research Network, hosted by the Halle Institute for Economic Research, with collaboration from various institutions, including the European Commission, European Bank for Reconstruction and Development, European Investment Bank, European Stability Mechanism, France Stratégie, the German Council of Economic Experts, the German Federal Ministry for Economic Affairs and Climate Action, and the Tinbergen Institute.

The dataset consists of micro-aggregated indicators derived from administrative balance sheet data from the 22 countries. Data providers are National Statistical Institutes (NSI) as well as researchers in National Central Banks and in selected academic institutions, which collect administrative firm-level data representative of the entire population of firms. All indicators are calculated using a harmonized data collection protocol, ensuring cross-country comparability.⁵ Using a distributed micro-data approach, the CompNet team calculates indicators across multiple levels of aggregation, such as country, macro-sector, industry,

⁵See Altomonte et al. (2018) for a discussion of cross-country comparability of CompNet, and Altomonte and di Mauro (2022) for an application of CompNet to derive empirical evidence on policies aimed at increasing overall productivity.

region (NUTS2), technology/knowledge category, and company age.⁶ At each aggregation level, there are nearly 600 variables organized into 6 main categories: finance, productivity, labor, competition, trade, and other areas. Each variable includes moments of the distribution, decompositions, and joint distributions, and CompNet also publishes transition matrices for selected variables. A detailed description of variables, country coverage, and data sources can be found in [CompNet \(2022\)](#). The CompNet dataset is publicly available on request for research purposes.

We employ the so-called “20e” version of the CompNet dataset, which is compiled by setting a harmonized reporting threshold for firms with at least 20 employees and re-weighted to align representativeness of sectors and size classes with Eurostat. This dataset contains information for twenty EU countries (BE, HR, CZ, DK, FI, FR, DE, HU, IT, LV, LT, MT, NL, PL, PT, RO, SK, SI, ES, and SE) plus one non-EU country (CH). Trade information is available for all those EU countries except BE, IT, LV, and ES. Data limitations thus constrain us to use a ‘EU16’ definition of the European Union, without BE, IT, LV, and ES, as well as EE, EL, BG, LU, CY, AT, and IE. The forced exclusion of IT and ES thus shifts the geographical barycenter of our sample away from the Mediterranean. Table A.1 in Appendix A reports the data coverage for the CompNet 20e dataset vis-à-vis Eurostat SBS and OCED ICIO.

The other dataset we use is the OECD Inter-Country Input-Output tables (ICIO). For our purposes, the tables map international trade flows across countries and sectors of economic activity covering 45 ISIC Rev. 4 2-digit sectors in 76 countries and the rest of the world. In particular, the tables provide bilateral exchanges of goods and services for the manufacturing sectors (divisions in Nace Rev. 2 section C) in the 16 countries of our sample that report trade variables in CompNet. This allows us to match each exporting country and sector with the aggregate exports to all other countries and sectors, including those in the EU and extra-EU that are not represented in CompNet. ICIO information is necessary to compute two critical variables later described in the paper, namely trade freeness and BRCA. However, while the OECD ICIO tables provide aggregate international flows, they do not feature micro trade data, especially export propensity (share of exporting firms), which we source from CompNet.⁷

3 Revealed Relative State of Technology: a Sufficient Statistics Approach

Our objective is to understand the extent to which a country’s export specialization across industries is driven by their technological prowess relative to foreign industries. Technological prowess is, however, hard to measure directly from the observed productivity of an industry as this is determined by many other concurring factors. To circumvent this problem, we apply the model by [Huang and Ottaviano \(2024\)](#), who modify the perfectly competitive general equilibrium approach [Dornbusch et al. \(1977, 1980\)](#) with a continuum of industries to allow them to feature monopolistic competition among heterogeneous firms as in [Mayer et al. \(2014\)](#). In the model export specialization is determined by the interplay of all the main sources of export specialization highlighted in the literature: relative state of technology ([Ricardo, 1817](#)); relative factor endowments ([Heckscher and Ohlin, 1933](#)); relative market size and relative market access ([Krugman, 1980](#)); relative competitive pressure and firm selection ([Melitz, 2003](#)). It thus provides a theory-consistent structural way to design a crosswalk between industries’ export performance and technological prowess controlling for other factors.

The model considers two countries, the country of interest (‘Home’) and a reference country (‘Foreign’), with given numbers of consumers L and L^* respectively, which we take as measures of their market sizes. In each country consumers are endowed with a unit of labor, which they supply inelastically so that L and L^* are also the countries’ labor endowments. Each consumer is also endowed with K/L or K^*/L^* units of capital depending on the country of residence so that K and K^* are the countries’ capital endowments. There is a continuum of industries indexed $z \in [0, 1]$, each of which supplies a set $\Omega(z)$ of horizontally differentiated

⁶See [Bartelsman et al. \(2004\)](#) and [Lopez-Garcia and di Mauro \(2015\)](#).

⁷For consistency, we also compute export intensity (exports on revenues) from CompNet although it is in principle observable in OECD ICIO.

varieties indexed $i \in \Omega(z)$. Labor and capital markets are perfectly competitive, whereas the product markets of the different industries are monopolistically competitive. While trade is free within countries, between them it faces iceberg frictions with $\tau(z) > 1$ units to be shipped for one unit to reach destination.

3.1 Preferences and Consumer Behavior

Consumers in both countries have identical preference. We focus on Home noting that symmetric expressions hold for Foreign. Individual utility is given by:

$$U = \int_0^1 b(z) \ln \left(\alpha \int_{i \in \Omega(z)} q_i^c(z) di - \frac{\gamma}{1-\delta} \int_{i \in \Omega(z)} q_i^c(z)^{1-\delta} di \right) dz,$$

where $q_i^c(z)$ denotes individual consumption of the differentiated product variety i in industry z and $b(z) > 0$ is the expenditure share of industry z with $\int_0^1 b(z) dz = 1$. The parameters $\gamma > 0$ and $\delta < 0$ capture the substitutability between varieties within each industry, which is a decreasing function of both.⁸ The budget constraint faced by a consumer is:

$$\int_0^1 \int_{i \in \Omega(z)} p_i(z) q_i^c(z) di dz = wL + rK/L,$$

where $p_i(z)$ is the price of variety i in industry z , while w and r are the wage of labor and the rental rate of capital. Solving the consumer's problem yields the following inverse individual demand for the differentiated varieties:

$$p_i(z) = p_{max}(z) - \frac{\gamma}{\lambda(z)} q_i^c(z)^{-\delta},$$

where $p_{max}(z) = \alpha/\lambda(z)$ is the 'choke price' above which demand is null, and $\lambda(z)$ is the marginal utility of income. The market demand for variety i in industry z is then given by:

$$q_i(z) = L \left(\frac{\gamma}{\lambda(z)} \right)^{\frac{1}{\delta}} (p_{max}(z) - p_i(z))^{-\frac{1}{\delta}}, \quad (1)$$

as L is the number of consumers in the home country. According to (1), the demand elasticity to price is a decreasing function of $p_{max}(z)$ and thus an increasing function of $\lambda(z)$. The choke price and the marginal utility of income can, therefore, be respectively taken as an inverse and a direct measures of industry z 's endogenous competitive pressure.

3.2 Production and Firm Behaviour

In all industries firms employ both labor and capital as inputs in a Cobb-Douglas production function such that the price of a unit input bundle in industry z evaluates to $\omega(z) = w^{\iota(z)} r^{1-\iota(z)} / [(1-\iota(z))^{\iota(z)} \iota(z)^{1-\iota(z)}]$ with $\iota(z) \in (0, 1)$. While labor intensity $\iota(z)$ differs across industries, within an industry it is the same for all firms, which nonetheless have different total factor productivity (TFP indexed by φ) or, equivalently different unit input requirement (UIR indexed by $c = 1/\varphi$). Then, a firm in industry z with UIR c has marginal cost $\omega(z)c$ as in [Corcos et al. \(2012\)](#). Given demand (1), the firm's maximized profit evaluates to

$$\pi_D(z, c) = -\delta L \left(\frac{\gamma}{\lambda(z)} \right)^{\frac{1}{\delta}} \left(\frac{p_{max}(z) - \omega(z)c}{1-\delta} \right)^{1-\frac{1}{\delta}} \quad (2)$$

for domestic sales and

$$\pi_X(z, c) = -\delta L^* \left(\frac{\gamma}{\lambda^*(z)} \right)^{\frac{1}{\delta}} \left(\frac{p_{max}^*(z) - \tau(z)\omega(z)c}{1-\delta} \right)^{1-\frac{1}{\delta}} \quad (3)$$

for export sales.

⁸For $\delta = -1$ utility is quadratic as in [Mayer et al. \(2014\)](#).

In each industry firms face a sunk entry cost incurred in units of its unit input bundle. Specifically, firms entering industry z pay an entry cost $\omega(z)f_E$ as in [Romalis \(2004\)](#) and [Bernard et al. \(2007\)](#). After entry, they draw their UIR from a Pareto distribution $G(z, c) = (c/C_M(z))^{k(z)}$, defined over the support $[0, C_M(z)]$ with shape parameter $k(z) > 1$ measuring the concentration of density towards its upper bound. Accordingly, $k(z)$ inversely captures the heterogeneity of the firm TFP distribution in industry z . The support defines firms at the technological frontier as those with $c = 0$ and all other firms as laggards, with their maximum and average distance from the frontier equal to $C_M(z)$ and $C_M(z)[k(z)/(k(z)+1)]$ respectively. Hence, larger $C_M(z)$ characterizes industries with larger maximum distance from the frontier and, for given $C_M(z)$, larger $k(z)$ characterizes those with larger average distance from the frontier. Analogously, firms entering industry z in Foreign draw their UIR from $G(z, c) = (c/C_M^*(z))^{k(z)}$, defined over the support $[0, C_M^*(z)]$ with the same $k(z) > 1$ as in Home.⁹ As also Foreign frontier firms feature $c = 0$, there is a common ‘global technological frontier’.

The fact that profits (2) and (3) are decreasing functions of c implies that, after all entrants have drawn their UIR, only those with c weakly lower than the domestic cutoff UIR $C_D(z) = p_{\max}(z)/\omega(z)$ make non-negative profits and sell in the domestic market. The others exit without producing. Among the survivors, only those with c weakly lower than the export cutoff UIR $C_X(z) = p_{\max}^*(z)/[\tau(z)\omega(z)] = [\omega(z)^*C_D^*(z)]/[\tau(z)\omega(z)] < C_D(z)$ also export to Foreign. Due to free entry in Home, the sum of expected profits from sales in the domestic and export markets equals the sunk entry cost:

$$\int_0^{C_D(z)} \pi_D(z, c) dG(z, c) + \int_0^{C_X(z)} \pi_X(z, c) dG(z, c) = \omega(z)f_E, \quad (4)$$

with an analogous expression holding for entrants in Foreign. After substituting the export cutoff in each country with its aforementioned expression in terms of the domestic cutoff in the other country, the free entry condition (4) for Home and its counterpart for Foreign can be solved together to obtain the domestic cutoffs. For Home, the solution is:

$$C_D(z) = \left[\frac{1 - \rho(z) \left(\frac{\omega^*(z)}{\omega(z)} \right)^{k(z)+1} \left(\frac{C_M^*(z)}{C_M(z)} \right)^{k(z)}}{1 - \rho(z)^2} \right]^{\frac{1}{k(z)+1}} C_D^A(z),$$

where $\rho(z) = \tau(z)^{-k(z)} \in [0, 1]$ is an index of trade freeness, while $C_D^A(z)$ is the autarkic domestic cutoff

$$C_D^A(z) = \left[C_M(z)^{k(z)} \frac{f_E}{\beta(z)L} \left(\frac{\alpha}{\gamma} \right)^{\frac{1}{\delta}} \right]^{\frac{1}{k(z)+1}}, \quad (5)$$

which corresponds to prohibitive trade frictions ($\rho(z) = 0$), where $\beta(z) = -\delta(1-\delta)^{(1-\delta)/\delta}k(z)B(k(z), 2-1/\delta)$ is a constant with $B(\cdot)$ denoting the Beta function.¹⁰ As $C_D^A(z)$ is an increasing function of $C_M(z)$ and a decreasing function of L , in autarky competitive pressure and thus firm selection are stronger the better the country’s state of technology and the larger its market size. In addition, for $\omega(z) = \omega^*(z)$ and $C_M(z) = C_M^*(z)$, $C_D(z)$ is a decreasing function of $\rho(z)$: freer trade (larger $\rho(z)$) raises competitive pressure and fosters firm selection (reduces $C_D(z)$). The larger $k(z)$, the larger the percentage fall in trade freeness $\rho(z)$ caused by a one percent rise in trade cost $\tau(z)$. This is due to the fact that with larger $k(z)$ there is larger share of marginal exporters that stop exporting when trade barriers increase.

3.3 Industry Productivity and Trade Performance

These expressions can be used to explore the relation between a Home industry’s average productivity and its trade performance at both the extensive and intensive margins. For productivity, [Huang and Ottaviano](#)

⁹As in [Eaton and Kortum \(2002\)](#), a common shape parameter across countries is needed for model tractability as it allows for closed-form aggregation.

¹⁰The Beta function is defined as $B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$.

(2024) define the TFPQ of Home industry z as the industry's output per unit of input bundle. They then use $\bar{\Phi}(z)$ and $\bar{\Phi}(z)^A$ to denote Home TFPQ with and without trade respectively. For trade, they use $\chi(z) = (C_X(z)/C_D(z))^{k(z)}$ to denote the industry's 'export propensity' defined as the fraction of its firms that export, and $\theta(z)$ to denote its 'export intensity' defined as the ratio of export revenues to total revenues.

Based on these definitions, they highlight three key properties. First, they show that export propensity $\chi(z)$, trade freeness $\rho(z)$, firms heterogeneity $k(z)$ and relative market size L/L^* regulate the relation between relative TFPQ with and without trade. In particular, relative TFPQ with trade can be decomposed as the product of relative TFPQ in autarky ('ex-ante component') and two factors ('ex-post components') capturing the parallel amplifying and dampening effects of trade liberalization on the translation of relative TFPQ without trade into relative TFPQ with trade:

$$\frac{\bar{\Phi}(z)}{\bar{\Phi}^*(z)} = \underbrace{\frac{\bar{\Phi}^A(z)}{\bar{\Phi}^{A*}(z)}}_{\text{ex-ante component}} \underbrace{\left(\frac{\frac{L}{L^*}\rho(z) + \left(\frac{\chi(z)}{\rho(z)}\right)^{\frac{k(z)+1}{k(z)}}}{\frac{L^*}{L}\rho(z) + \left(\frac{\chi(z)}{\rho(z)}\right)^{-\frac{k(z)+1}{k(z)}}} \right)^{\frac{k(z)}{k(z)+1}}}_{\text{ex-post amplifying component} > 1} \underbrace{\frac{\rho(z)\chi(z)^{-1} + \frac{L^*}{L}\rho(z)}{\rho(z)^{-1}\chi(z) + \frac{L}{L^*}\rho(z)}}_{\text{ex-post dampening component} < 1}, \quad (6)$$

where $\bar{\Phi}^A(z)/\bar{\Phi}^{A*}(z) = C_D^{A*}(z)/C_D^A(z)$ is the industry's relative TFPQ in autarky. Which ex-post component prevails cannot be determined theoretically but only empirically. Intuitively, in industries with higher export propensity $\chi(z)$, firm selection into exporting is weaker. This reduces the productivity premium of exporters relative to non-exporters and thus industry TFPQ at the extensive margin. On the other hand, industries with higher $\chi(z)$, also have a larger share of exporters. As these are more productive than non-exporters, that raises industry TFPQ.¹¹

Second, export intensity $\theta(z)$, trade freeness $\rho(z)$, firm heterogeneity $k(z)$ and relative market size L/L^* regulate the relation between relative TFPQ in autarky and relative unit input prices with trade:

$$\frac{\bar{\Phi}^A(z)}{\bar{\Phi}^{A*}(z)} = \frac{\omega(z)}{\omega^*(z)} [\rho(1 - \theta(z)) + \rho^{-1}\theta(z)]^{\frac{1}{k(z)+1}} \left(\frac{L}{L^*} \right)^{\frac{1}{k(z)+1}}. \quad (7)$$

Third and last, export intensity $\theta(z)$, trade freeness $\rho(z)$ and firm heterogeneity $k(z)$ determine the relation between relative state of technology and relative unit input prices with trade:

$$\frac{C_M^*(z)}{C_M(z)} = \left(\frac{\omega(z)}{\omega^*(z)} \right)^{\frac{k(z)+1}{k(z)}} [\rho(1 - \theta(z)) + \rho^{-1}\theta(z)]^{\frac{1}{k(z)}}. \quad (8)$$

This expression defines the theory-based index of revealed Relative State of Technology (RST).

The practical relevance of these properties comes from the fact that relative autarkic TFPQ ($\bar{\Phi}^A(z)/\bar{\Phi}^{A*}(z)$) is unobservable in the absence of data on a counterfactual situation without trade, while relative state of technology ($C_M^*(z)/C_M(z)$) would remain unobservable even if those data were available since $C_D^A(z)$ depends not only on $C_M(z)$ but also on L as already discussed. What (7) and (8) allow one to do is to use the observed export intensity $\theta(z)$, trade freeness $\rho(z)$, firm heterogeneity $k(z)$, relative market size L/L^* and relative unit input price $\omega(z)/\omega^*(z)$ to compute the unobserved relative autarkic TFPQ and state of technology without having to calibrate the entire model. Huang and Ottaviano (2024) call this a 'sufficient statistics approach'.

¹¹While a precise quantification of the ex-post amplifying and dampening components of TFPQ lies beyond the scope of the present analysis, additional results in di Mauro et al. (2025) show that the latter dominates in most sectors, except 20, 21, 26, 28, 29, and 30. As we will see, most of these sectors also exhibit BRCA and experience an increase in TFP from autarky to trade primarily because of an ineffective ex-post dampening component, rather than a strong ex-post amplifying one. This suggests that sectors with BRCA may be relatively insulated from trade-related consequences that could otherwise erode their productivity advantages.

4 The Technological Prowess of Europe

We are now ready to apply the sufficient statistics approach by [Huang and Ottaviano \(2024\)](#) to investigate the extent to which the EU's export specialization across industries is driven by its technological prowess relative to foreign industries, with special emphasis on its two biggest western economies (France and Germany) and its biggest eastern economy (Poland). We also test whether the relationship between export performance and technological advantage differs across destinations within and outside the single market, and how it evolves over time. As a measure of export specialization, we use the traditional index of 'revealed' comparative advantage ([Balassa, 1965](#)), henceforth called BRCA. For an industry in a given country of interest, the corresponding BRCA is traditionally defined as its share of national exports divided by the industry's share of exports in the Rest of the World (RoW). If the value of its BRCA is larger than 1, the country's relatively stronger export specialization in the industry 'reveals' its comparative advantage in that industry. Analogously, a value of its BRCA smaller than 1 reveals the country's relatively weaker export specialization in the industry.

4.1 Sufficient Statistics

Following [Huang and Ottaviano \(2024\)](#), for a given year relative unit input prices $\omega(z)/\omega^*(z)$ are retrieved from the Penn World Table 10.01. For the country of interest we compute the wage rate as real GDP multiplied by the labor share and divided by total employment. The labor share $\iota(z)$ is calculated as the industry-level median of firm labor cost relative to value added for each industry from CompNet.¹² For the rental rate of capital, we take the real internal rate of return. Measures for the rest of the world (RoW) are obtained as weighted average across all countries without missing data, with the wage rate weighted by employment and the rental rate weighted by capital stock. These pieces of information allow us to evaluate the country-industry price of a unit input bundle in industry z , which we defined as $\omega(z) = w^{\iota(z)} r^{1-\iota(z)} / [w^*(z)^{\iota(z)} r^*(z)^{1-\iota(z)}]$.

As for relative market size and trade freeness, we compute them in a model-consistent way as

$$\frac{L(z)}{L^*(z)} = \frac{1 - \theta(z)}{\theta(z)} \chi(z)^{\frac{k(z)+1}{k(z)}} \rho(z)^{-\frac{1}{k(z)}}$$

and, following [Head and Ries \(2001\)](#), as

$$\rho(z) = \sqrt{\frac{Exp(z)}{S(z)} \frac{Exp^*(z)}{S^*(z)}}, \quad (9)$$

where $Exp(z)$ and $S(z)$ are industry z 's exports and domestic sales of the country of interest while $Exp^*(z)$ and $S^*(z)$ are the analogues for the RoW. These are lifted from the OECD Inter Country Input Output tables. For the country of interest, $Exp(z)$ is equal to total industry output $Out(z)$ minus domestic sales: $Exp(z) = Out(z) - S(z)$. For the RoW, we proceed as follows. We use superscript $^\circ$ to label a RoW country, such that $Exp^\circ(z) = Out^\circ(z) - S^\circ(z)$. We then sum $Exp^\circ(z)$ and $S^\circ(z)$ across all RoW countries to obtain $Exp^*(z) = \sum_o Exp^\circ(z)$ and $S^*(z) = \sum_o S^\circ(z)$. This implies that

$$\frac{Exp^*(z)}{S^*(z)} = \frac{\sum_o S^\circ(z) \frac{Exp^\circ(z)}{S^\circ(z)}}{\sum_o S^\circ(z)}$$

is a weighted average of $Exp^\circ(z)/S^\circ(z)$ across RoW countries with weights given by their shares $S^\circ(z)/\sum_o S^\circ(z)$ of total RoW domestic sales. Hence, $Exp^*(z)/S^*(z)$ is specific to the country of interest because the RoW includes all countries except that country. In the case of the RoW, sales between its composing countries are classified as 'exports' rather than 'domestic sales' within an integrated RoW market. We apply an analogous consistent aggregation to EU countries when considering the entire European Union as 'country of interest'.

¹²However, when retrieving the EU aggregate, we consider the average industry labor share across countries.

This allows us to interpret the resulting trade freeness and BRCA for the aggregates as weighted averages of the corresponding measures for the underlying countries.

Finally, firm heterogeneity $k(z)$ is obtained by averaging across countries the skewness of the within-industry firm distribution of TFP 3 in CompNet, which results from the estimation of TFP following [Akerberg et al. \(2015\)](#). As reported in Table B.1 of Appendix B, the average skewness of the TFP distribution, pooling all industries, countries, and years in our CompNet sample, is 1.154. This is in the ballpark of existing estimates. Depending on the specific log-log estimation method applied to the cumulative distribution function of the inverse of estimated TFP, [Huang and Ottaviano \(2024\)](#) found a comparable measure for China ranging from 0.887 to 1.429. [Head et al. \(2014\)](#) estimated a Pareto shape parameter of 1.367 using Chinese exports to Japan for the year 2000. While we could not find estimates of the same parameter for the US, the Bureau of Labor Statistics and the US Census Bureau report the interquartile range of gross output per unit of combined inputs for 86 four-digit NAICS manufacturing industries. For the period of our analysis (2010-2018), the mean interquartile range across industries and years is 0.520. The average across industries, countries, and years for a comparable measure in CompNet, which however excludes capital costs from combined inputs, was 0.357.

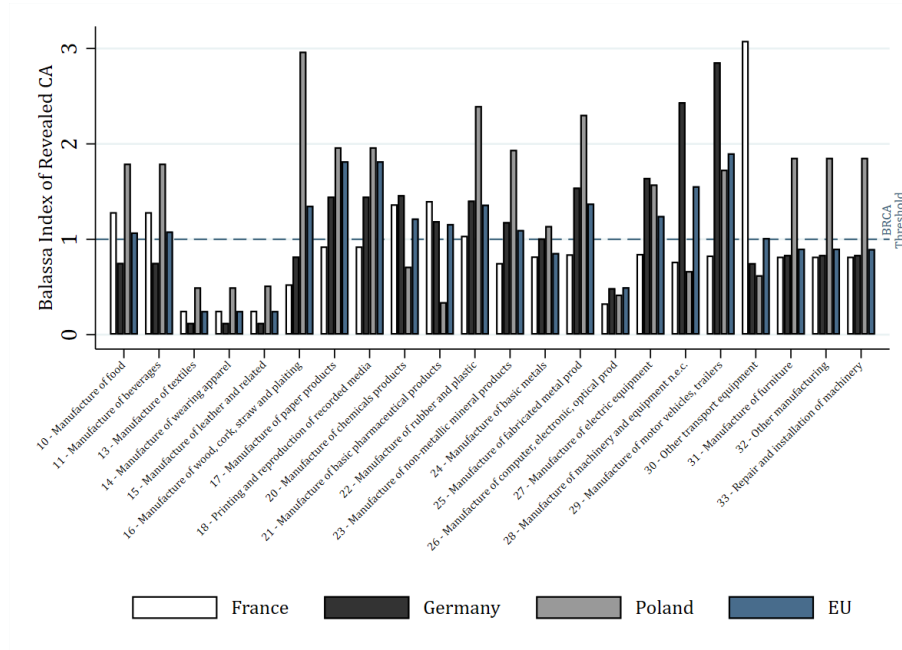
To summarize, for a given year, export propensity $\chi(z)$, export intensity $\theta(z)$, relative unit input price $\omega(z)/\omega^*(z)$, relative market size $L(z)/L^*(z)$ and trade freeness $\rho(z)$ vary across countries and industries, while firm heterogeneity $k(z)$ varies across industries. Summary statistics for France, Germany, Poland, and the European Union are provided in Appendix B.

4.2 Export Specialization and State of Technology

The BRCA for France, Germany, Poland, and the European Union with respect to the rest of the world is illustrated in Figure 1. The figure shows that the EU has a BRCA in thirteen out of twenty-two industries: 10 (Food), 11 (Beverages), 16 (Wood), 17 (Paper), 18 (Printing and recorded media), 20 (Chemical), 21 (Pharmaceutical), 22 (Rubber and plastic), 23 (Non-metallic mineral), 25 (Fabricated metal), 27 (Electric equipment), 28 (Machinery) and 29 (Motor vehicles). France and Germany are remarkably different in their actual export performance. France has a BRCA in sectors 10, 11, 20, 21 and (marginally) 22. Differently from the entire EU, France has a strong BRCA in 30 (Other transport equipment; likely driven by Airbus). While Germany shares with France its BRCA in sectors 20, 21 and 22, it does not share its BRCA in sectors 10, 11 and 30. It has, however, a BRCA in several other sectors: 17, 18, 23, 25, 27, 28 and 29. This also holds for Poland. While this country has a BRCA in 10 and 11 as France, the pattern of its BRCA mirrors the German one in 16, 17, 18, 22, 23, 25, 27 and 29, possibly due to its involvement in Germany's value chains. When comparing the three EU countries, what sets a country apart from the others is its BRCA in the following industries: 30 (Other transport equipment) for France; 28 (Machinery) for Germany; 16, 24 (Basic metals), 31 (Furniture), 32 (Other manufacturing) and 33 (Repair and installation) for Poland. In all three countries and the EU as a whole, strong revealed comparative disadvantage characterizes not only relative low-tech sectors such as 13 (Textiles), 14 (Wearing apparel) and 15 (Leather), but also a high-tech sector such as 26 (Computer).

All this is very much in line with what one would expect, but how does it compare with the revealed relative state of technology (RST)? Figure 2 shows the results obtained from computing the RST of French sectors based on expression (8). While France has a better state of technology than the world average in virtually all sectors, the RST pattern across sectors is quite different from the BRCA one. Only sectors 20 and 21 exhibit alignment in relative position across both BRCA and RST. The remaining French sectors display divergent patterns. Specifically, sectors 13, 14, 15, 23, 24, 25, 26 and 32 show particularly unfavorable relative BRCA values with respect to their corresponding RST. In contrast, sector 30 - the sector with the highest BRCA - is relatively much more successful in terms of BRCA than RST.

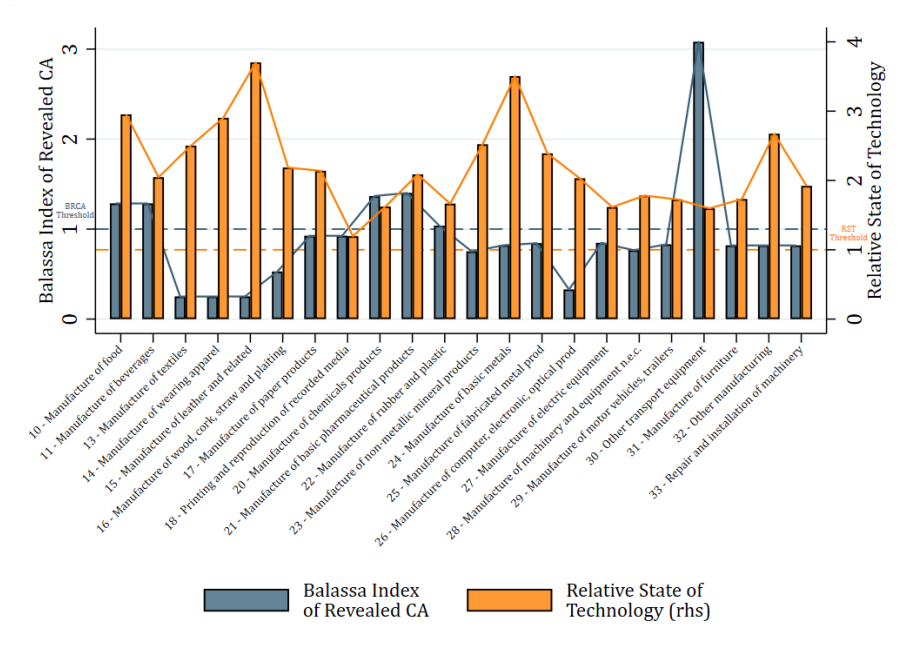
Figure 1: Balassa Revealed Comparative Advantage in France, Germany, Poland, and the EU



Source: CompNet, OECD ICIO, and calculations by the authors.

Note: The dashed blue line represents the threshold for the *Balassa Index of Revealed CA* to indicate specialization. The OECD ICIO tables consolidate sectors 10, 11, 13, 14, 15, 31, 32, and 33 into broader categories. For these sectors, the figure for the corresponding category is reported. Average from 2010 to 2018.

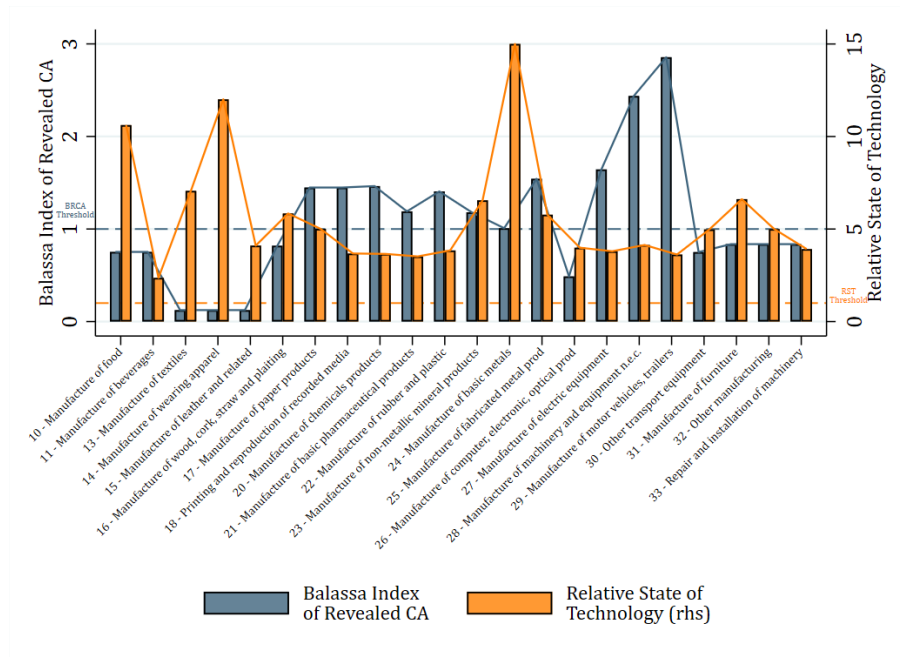
Figure 2: Balassa Revealed Comparative Advantage and Relative State of Technology in France



Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Note: *Balassa Index of Revealed CA* is the the ratio of an industry's share of exports for France to its share in world exports. *Relative State of Technology* is computed like in equation (8). *Balassa Index of Revealed CA* and *Relative State of Technology* are represented on the left-hand and right-hand axes, respectively. The dashed blue line represents the threshold for the *Balassa Index of Revealed CA* to indicate specialization. The dashed orange line represents the threshold for the *Relative State of Technology* to indicate technological advantage. The OECD ICIO tables consolidate sectors 10, 11, 13, 14, 15, 31, 32, and 33 into broader categories. For these sectors, the *Balassa Index of Revealed CA* for the corresponding category is reported. Average from 2010 to 2018.

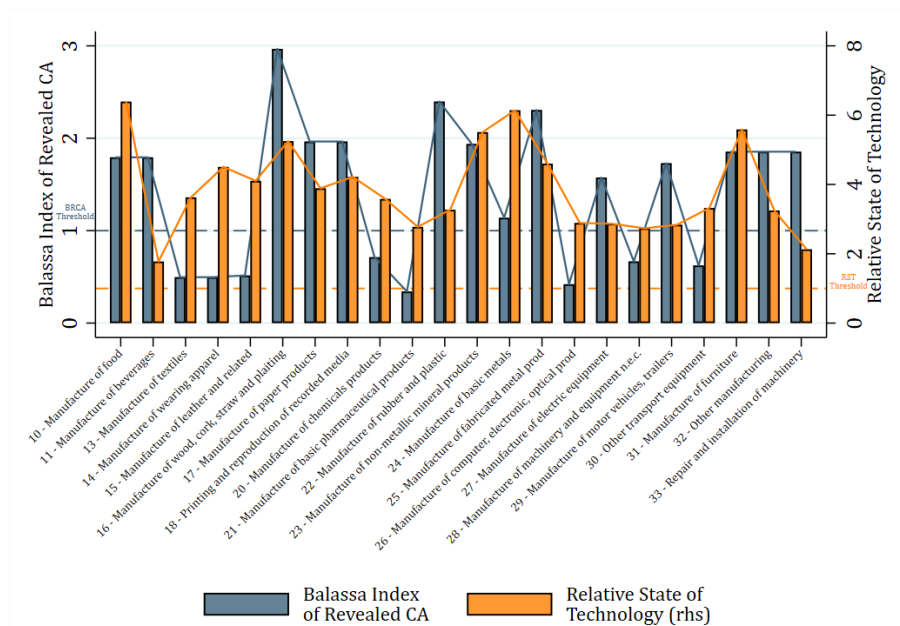
Figure 3: Balassa Revealed Comparative Advantage and Relative State of Technology in Germany



Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Note: *Balassa Index of Revealed CA* is the the ratio of an industry's share of exports for Germany to its share in world exports. *Relative State of Technology* is computed like in equation (8). *Balassa Index of Revealed CA* and *Relative State of Technology* are represented on the left-hand and right-hand axes, respectively. The dashed blue line represents the threshold for the *Balassa Index of Revealed CA* to indicate specialization. The dashed orange line represents the threshold for the *Relative State of Technology* to indicate technological advantage. The OECD ICIO tables consolidate sectors 10, 11, 13, 14, 15, 31, 32, and 33 into broader categories. For these sectors, the *Balassa Index of Revealed CA* for the corresponding category is reported. Average from 2010 to 2018.

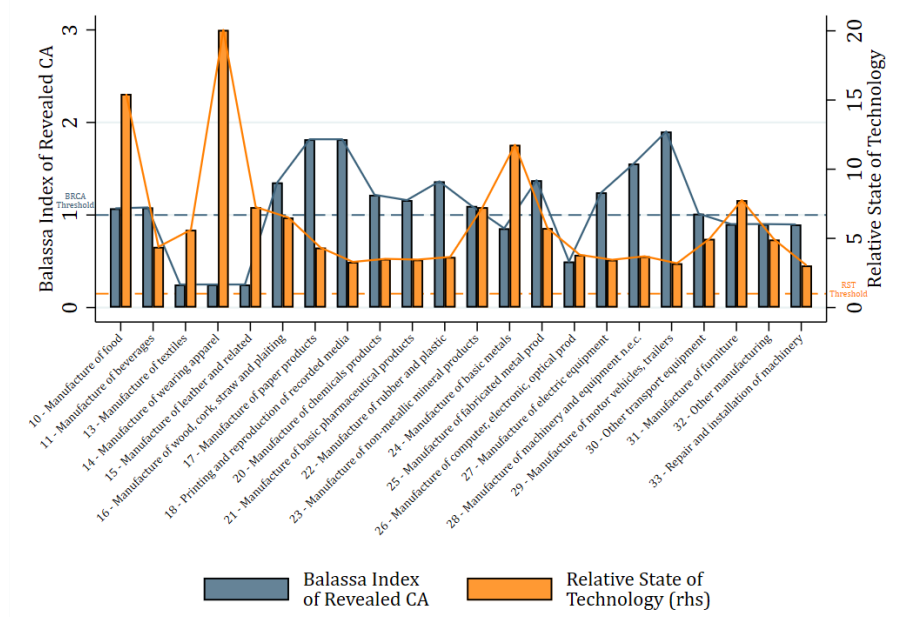
Figure 4: Balassa Revealed Comparative Advantage and Relative State of Technology in Poland



Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Note: *Balassa Index of Revealed CA* is the the ratio of an industry's share of exports for Poland to its share in world exports. *Relative State of Technology* is computed like in equation (8). *Balassa Index of Revealed CA* and *Relative State of Technology* are represented on the left-hand and right-hand axes, respectively. The dashed blue line represents the threshold for the *Balassa Index of Revealed CA* to indicate specialization. The dashed orange line represents the threshold for the *Relative State of Technology* to indicate technological advantage. The OECD ICIO tables consolidate sectors 10, 11, 13, 14, 15, 31, 32, and 33 into broader categories. For these sectors, the *Balassa Index of Revealed CA* for the corresponding category is reported. Average from 2010 to 2018.

Figure 5: Balassa Revealed Comparative Advantage and Relative State of Technology in the EU



Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Note: *Balassa Index of Revealed CA* is the ratio of an industry's share of exports for the EU to its share in world exports. *Relative State of Technology* is computed like in equation (8). *Balassa Index of Revealed CA* and *Relative State of Technology* are represented on the left-hand and right-hand axes, respectively. The dashed blue line represents the threshold for the *Balassa Index of Revealed CA* to indicate specialization. The dashed orange line represents the threshold for the *Relative State of Technology* to indicate technological advantage. The figures for the EU are derived by aggregating components (exports, export intensity, export propensity, trade freeness, and relative unit input prices) across the EU countries included in our sample: HR, CZ, DK, FI, FR, DE, HU, LT, MT, NL, PL, PT, RO, SK, SI, and SE. The OECD ICIO tables consolidate sectors 10, 11, 13, 14, 15, 31, 32, and 33 into broader categories. For these sectors, the *Balassa Index of Revealed CA* for the corresponding category is reported. Average from 2010 to 2018.

Figure 3 reports the corresponding results for Germany. As for France, there are sectors with relatively high BRCA (such as 27, 28, and 29) that are not 'backed' by relatively high RST. For sectors 24, 14, 10 and 13, the divergence between BRCA and RST is particularly high, implying that relative export performance has rather little to do with the relative state of technology.

A similar pattern is observed also for Poland in Figure 4, where sectors 13, 14, 15, 20, 21, 26 and 30 exhibit sustained RST despite subdued BRCA. However, among sectors with strong BRCA, only 11, 22, and 33 are associated with relatively low RST.

When considering the EU as a whole in Figure 5, a picture similar to the German one emerges. Specifically, sectors 29, 18, 17, 28, and 27 are relatively much more successful in terms of BRCA than RST. In contrast, sectors 14, 10 and 24 show particularly unfavorable BRCA relative to the corresponding RST.

Notably, the highest BRCA of France in sector 30 (Other transport equipment) is associated with an RST of 1.60, which is low compared to the largest RST in France (3.70 for sector 15 Leather). It is also lower than the RST of the very same sector 30 in Germany (4.98) and in the EU on average (3.20). This divergence between French BRCA and RST is likely driven by Airbus and may be explained in terms of the important role played by scale economies in aircraft production.

Going beyond visual inspection, the first row of Table 2 reports the within-country correlation between industry rankings based on the BRCA and the RST of the five largest economies in our sample and the group of Central, Eastern, and Southeastern European (CESEE) countries, confirming the lack of significant correlation (Germany, Netherlands, Poland, Sweden) or even the presence of some negative correlation (France and CESEE).¹³ In the full EU sample (see row 1 column 4 of Table 1), the rankings of industries

¹³Rankings are inversely proportional to the magnitude of the underlying variable, with rank 1 indicating the highest value.

based on BRCA and RST exhibit significant negative correlation.¹⁴ While column 4 of Table 2 suggests that the discrepancy between BRCA and RST is less severe for Poland than for Germany and France, column 6 shows that the discrepancy in CESEE countries is more pronounced than in the full sample.¹⁵

Table 1: Pairwise Correlations between Productivity Measures

	(1) Ricardian CA $\frac{\Phi^A(z)}{\Phi^{A*}(z)}$	(2) TFPQ $\frac{\Phi(z)}{\Phi^*(z)}$	(3) CompNet Productivity	(4) Balassa Revealed CA $\frac{\frac{\sum_z Exp(z)}{\sum_z ExpWorld(z)}}{ExpWorld(z)}$
Relative State of Technology $\frac{C^*_M(z)}{C_M(z)}$	0.189*** (0.017)	0.347*** (0.016)	-0.041** (0.016)	-0.034*** (0.010)
R^2 Obs.	0.92 2,576	0.93 2,576	0.93 2,576	0.97 2,576
Ricardian CA $\frac{\Phi^A(z)}{\Phi^{A*}(z)}$		0.752*** (0.015)	0.044** (0.019)	-0.056*** (0.012)
R^2 Obs.		0.96 2,576	0.93 2,576	0.97 2,576
TFPQ $\frac{\Phi(z)}{\Phi^*(z)}$			0.045** (0.019)	-0.060*** (0.012)
R^2 Obs.			0.93 2,576	0.97 2,576
CompNet Productivity				-0.018 (0.013)
R^2 Obs.				0.97 2,576

Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Notes: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table reports the coefficients from separate regressions of the column variable on each row variable. A unit of observation is a year-country-industry combination. *Balassa Revealed CA* is the the ratio of an industry's share of exports for a given country to its share in world exports. *TFPQ* and *Ricardian CA* are computed like in equations (6) and (7), respectively. *Relative State of Technology* is computed like in equation (8). *CompNet Productivity* is the labor productivity of the median firm in the industry, measured as value added per employee. For each measure, the value corresponds to the ranking of the industry within a given country and year. All regressions include country-industry and year fixed effects. Estimations for HR, CZ, DK, FI, FR, DE, HU, LT, MT, NL, PL, PT, RO, SK, SI, and SE between 2010 and 2018.

As a sanity check, Table 1 looks at the rank correlations between relative TFPQ with trade ($\bar{\Phi}(z)/\bar{\Phi}^*(z)$), relative TFPQ without trade ($\bar{\Phi}^A(z)/\bar{\Phi}^{A*}(z)$),¹⁶ BRCA, and labor productivity, with labor productivity computed from CompNet data as value added per employee for the median firm. To avoid confusion, we call relative TFPQ with trade simply TFPQ and relative TFPQ without trade Ricardian (i.e. productivity-based) comparative advantage (CA). The table shows the alignment of CompNet productivity with both TFPQ and Ricardian CA, and the misalignment of the various productivity measures with BRCA.

¹⁴Estimation in Table 1 includes combined country-industry fixed effects, capturing the extent of misalignment between RST and BRCA due to factors that remain largely unchanged over the analysis period. These may include structural policies targeting export activity in specific industries, institutional features, positions within global value chains, and geographical or spatial characteristics. All these factors deserve deeper analysis, which we highlight as a promising avenue for future research. The results are generally robust to clustering standard errors by country, except some loss of significance for CompNet productivity in column 3.

¹⁵Table 2 also documents the limited correlation between industry rankings based on BRCA and the other productivity measures discussed in Section 4.1, as well as with rankings based on productivity estimates available in the CompNet dataset.

¹⁶Computed according to expressions (6) and (7), respectively.

Table 2: Pairwise Correlations between Export Specialization and Productivity Measures, by Country

	(1)	(2)	(3)	(4)	(5)	(6)
Balassa Revealed	Germany	France	Netherlands	Poland	Sweden	CESEE
CA $\frac{\frac{Exp(z)}{\sum_z Exp(z)}}{\frac{Exp^{World}(z)}{\sum_z Exp^{World}(z)}}$						
Relative State of Technology	-0.055	-0.106**	0.047	-0.016	0.036	-0.045***
$\frac{C^*_M(z)}{C_M(z)}$	(0.035)	(0.049)	(0.060)	(0.059)	(0.031)	(0.015)
R^2	0.97	0.95	0.96	0.94	0.98	0.97
Obs.	198	196	108	186	169	1,363
Ricardian CA	-0.049	-0.123*	-0.034	0.021	-0.013	-0.076***
$\frac{\Phi^A(z)}{\Phi^{A*}(z)}$	(0.055)	(0.072)	(0.101)	(0.083)	(0.041)	(0.017)
R^2	0.97	0.95	0.96	0.94	0.98	0.97
Obs.	198	196	108	186	169	1,363
TFPQ	-0.053	-0.146*	0.066	-0.033	0.028	-0.069***
$\frac{\Phi(z)}{\Phi^*(z)}$	(0.055)	(0.077)	(0.118)	(0.080)	(0.052)	(0.015)
R^2	0.97	0.95	0.96	0.94	0.98	0.97
Obs.	198	196	108	186	169	1,363
CompNet Productivity	-0.121	-0.096	0.002	0.099	-0.040	-0.012
	(0.113)	(0.066)	(0.065)	(0.090)	(0.039)	(0.017)
R^2	0.97	0.95	0.96	0.94	0.98	0.97
Obs.	198	196	108	186	169	1,363

Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Notes: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table reports the coefficients from separate regressions of the *Balassa Revealed CA* on each row variable for the five largest economies in our sample (columns 1-5) and CESEE countries (column 6). A unit of observation is a year-country-industry combination. *Balassa Revealed CA* is the ratio of an industry's share of exports for a given country to its share in world exports. *TFPQ* and *Ricardian CA* are computed like in equations (6) and (7), respectively. *Relative State of Technology* is computed like in equation (8). *CompNet Productivity* is the labor productivity of the median firm in the industry, measured as value added per employee. For each measure, the value corresponds to the ranking of the industry within a given country and year. All regressions include industry fixed effects, but the ones in column 6 for CESEE countries that include country-industry and year fixed effects. CESEE countries are HR, CZ, HU, LT, PL, RO, SK, and SI. Non-CESEE countries are DK, FI, FR, DE, MT, NL, PT, and SE. Estimations are for years between 2010 and 2018.

As some of the reports on European competitiveness discussed in the Introduction express concern about the loss of technological advantage compared to China and the US, it is interesting to look at the EU's BRCA and RST with respect to those two countries. Indeed, for optimizing policy efforts aimed at bolstering the EU's competitiveness, it is essential to determine whether the relative export performance with respect to major global competitors stems primarily from underlying technological strengths or weaknesses, rather than from additional factors influencing specialization beyond structural productivity, which are examined in greater analytical detail in Table 4. Figure C.1 in Appendix C compares the EU's export specialization and technological advantage specifically against China (panel C.1a) and the US (panel C.1b) by replacing the RoW with either of them in the computation of trade freeness and relative unit input prices. As to be expected, when compared with the US rather than the RoW, the RST of the EU weakens in several industries, with $RST < 1$ in eleven industries (18, 20, 21, 22, 24, 26, 27, 28, 29, 30, 33) including Motor vehicles (29). By contrast, the EU holds a BRCA with respect to the US in all industries except 26 (Computer) and 30 (Other transport equipment; despite Airbus).

When compared with China instead, the EU remains technologically more advanced ($RST > 1$) across all sectors, but its BRCA is consistently lower than with respect to the RoW. Relative to China rather than the RoW, the EU even loses its revealed comparative advantage in some sectors, such as 27.

4.3 Single Market Versus Rest of the World

To deepen our investigation of the underpinnings of the EU's competitiveness, we further leverage our framework to pursue the following objectives. First, we assess whether differences exist in the divergence between technology and trade patterns within the single market relative to exchanges with the rest of the

world, and whether such differences have evolved over time. Second, in sections 4.4 and 4.5, we evaluate the factors that may contribute to driving this misalignment. Table 3 breaks down the results on the first row of Table 2 by export destination and time period, indicating that, as expected, the misalignment between export specialization and RST is typically more pronounced for destinations outside the EU. Throughout the European debt crisis and the subsequent recovery, its magnitude did not change significantly for overall and within-EU exchanges, but there is some evidence that the discrepancy increased over time when exporting outside the EU. This trend was driven primarily by Germany, for which, by the end of the period, the negative correlation with RST in extra-EU flows had become more substantial than in France, as well as by CESEE countries. The lack of resemblance between the rankings for RST and BRCA being more relevant for extra- than within-EU destinations reflects a different responsiveness to the underlying factors. This is illustrated by the results in Table 5 showing that, compared to aggregate flows, within the EU, country–industry relative market size is more relevant, whereas relative unit input prices, firm heterogeneity, and trade freeness play a stronger role outside the EU.

Table 3: Correlation between BRCA and RST within and outside the EU through Time, by Country

Balassa Revealed $\frac{C^*_M(z)}{C_M(z)} \times \frac{\sum_z Exp^{c \rightarrow d}(z)}{\sum_z Exp^{World}(z)}$	All Countries			Germany			France			CESEE		
	All	In EU	Ex EU	All	In EU	Ex EU	All	In EU	Ex EU	All	In EU	Ex EU
CA	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Relative State of Technology $\frac{C^*_M(z)}{C_M(z)}$	-0.033*** (0.011)	0.002 (0.011)	-0.038*** (0.011)	-0.028 (0.040)	-0.018 (0.028)	-0.089** (0.035)	-0.134** (0.055)	-0.001 (0.051)	-0.106*** (0.036)	-0.042** (0.016)	-0.015 (0.016)	-0.037** (0.017)
Relative State of Technology $\frac{C^*_M(z)}{C_M(z)} \times 2013-2015$	-0.004 (0.009)	0.003 (0.009)	-0.018** (0.009)	-0.025 (0.033)	0.007 (0.023)	-0.047 (0.029)	0.045 (0.043)	0.044 (0.040)	0.036 (0.029)	-0.011 (0.013)	-0.000 (0.013)	-0.015 (0.013)
Relative State of Technology $\frac{C^*_M(z)}{C_M(z)} \times 2016-2018$	0.000 (0.009)	-0.008 (0.009)	-0.021** (0.009)	-0.051 (0.033)	0.004 (0.023)	-0.058** (0.029)	0.046 (0.045)	0.029 (0.042)	0.027 (0.030)	-0.001 (0.013)	-0.010 (0.013)	-0.022* (0.013)
R^2	0.97	0.97	0.97	0.97	0.99	0.98	0.95	0.96	0.98	0.97	0.97	0.97
Obs.	2,576	2,576	2,576	198	198	198	196	196	196	1,363	1,363	1,363

Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Notes: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table reports the coefficients from regressions of the *Balassa Revealed CA* on the *Relative State of Technology* and time interactions for all countries in our sample (columns 1-3), Germany (columns 4-6), France (columns 7-9), and CESEE countries (columns 10-12). A unit of observation is a year-country-industry combination. *Relative State of Technology* is computed like in equation (8). *Balassa Revealed CA* is computed like $[Exp^{c \rightarrow d}(z) / \sum_z Exp^{c \rightarrow d}(z)] / [Exp^{World}(z) / \sum_z Exp^{World}(z)]$ where $Exp^{c \rightarrow d}(z)$ are total exports in industry z from country c to destination d , and d is RoW in columns 1, 4, 7, and 10, the rest of the EU in columns 2, 5, 8, and 11, and RoW outside of the EU in columns 3, 6, 9, and 12. Baseline for 2010-2012. *2013-2015* and *2016-2018* are dummies for the respective years. For each measure, the value corresponds to the ranking of the industry within a given country and year. All regressions include industry and year fixed effects, but the ones in columns 1-3 for all countries and 10-12 for CESEE countries that include country-industry and year fixed effects. CESEE countries are HR, CZ, HU, LT, PL, RO, SK, and SI. Non-CESEE countries are DK, FI, FR, DE, MT, NL, PT, and SE.

The statistically significant misalignment with RST in extra-EU exports being reflected in the overall exports for CESEE countries, but not for Germany, appears puzzling given that within-EU exports generally account for a considerably higher share of total exports for those countries than for the rest of the EU. One possible explanation lies in the fact that the correlation between the rankings of within- and extra-EU BRCA is stronger for CESEE countries (0.906) than for non-CESEE countries (0.847) and for Germany (0.789), making it more likely that, although with different intensity, the industries driving the divergence with respect to the RST overlap across destinations.

4.4 When Do RST and BRCA Differ More?

Table 4 deepens the analysis of the drivers of RST and BRCA. Columns 1 and 2 rank regress the RST and the BRCA on the items the model predicts those elements should covary with: relative input prices ($\omega(z)/\omega^*(z)$), firm heterogeneity (inversely measured by $k(z)$), trade freeness ($\rho(z)$), relative market size (L/L^*), and export intensity ($\theta(z)$). For each element or item in industry z , we consider the ranking of the industry in a given country and year with higher rank associated with smaller values. Higher ranked RST (column 1) is associated with higher ranked (hence smaller) relative unit input prices, with less firm heterogeneity (lower ranked, hence larger $k(z)$), lower ranked (hence more) trade freeness, and higher ranked

(hence lower) export intensity.¹⁷

Column 3 compares RST (column 1) and BRCA (column 2) by rank regressing RST-BRCA on the key items they are associated with across countries, industries and years, including both country-industry and year fixed effects. It shows that, within a country-industry, after controlling for time-specific shocks or trends that affect all countries-industries similarly, years with larger rank difference between RST and BRCA are associated with higher ranked $\omega(z)/\omega^*(z)$ (i.e. lower relative unit input price) as well as lower ranked $k(z)$ (i.e. less heterogeneity as captured by larger $k(z)$), lower ranked $\rho(z)$ (i.e. more trade freeness) and lower ranked L/L^* (i.e. larger relative market size).¹⁸ In other words, lower unit input price and less heterogeneity as well as more trade freeness and larger relative market size boost a country-industry's relative export performance with respect to its relative state of technology.

Table 4: Estimation of Components of Productivity Measures

	(1) Relative State of Technology $\frac{C_M^*(z)}{C_M(z)}$	(2) Balassa Revealed CA $\frac{\frac{Exp(z)}{\sum_z Exp(z)}}{\frac{Exp^*(z)}{\sum_z Exp^*(z)}}$	(3) RST - BRCA
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	0.142*** (0.012)	0.011 (0.007)	0.135*** (0.015)
Firms Heterogeneity $k(z)$	-0.401*** (0.016)	-0.014 (0.010)	-0.397*** (0.020)
Trade Freeness ρ	-0.466*** (0.031)	0.146*** (0.019)	-0.594*** (0.039)
Relative Market Size $\frac{L}{L^*}$		-0.021 (0.013)	-0.094*** (0.027)
Export Intensity $\theta(z)$	0.376*** (0.021)		
R^2	0.92	0.97	0.94
Obs.	2,576	2,576	2,576

Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Notes: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. A unit of observation is a year-country-industry combination. *Relative State of Technology* is computed like in equation (8). *Balassa Revealed CA* is the the ratio of an industry's share of exports for a given country to its share in world exports. For each measure, the value corresponds to the ranking of the industry within a given country and year. *RST-BRCA* is the difference between the respective rankings in columns (1) and (2). All regressions include country-industry and year fixed effects. Estimations for HR, CZ, DK, FI, FR, DE, HU, LT, MT, NL, PL, PT, RO, SK, SI, and SE between 2010 and 2018.

These findings confirm the empirical relevance of all the disturbances that, according to the model, can lead to the misalignment of BRCA and RST.

4.5 Drivers of Divergence in the Single Market Versus the Rest of the World

Section 4.3 has established that the divergence between RST and BRCA is heterogeneous across EU member states, export destinations, and over time. From a policy perspective, it is crucial to investigate how the relative importance of the factors identified in Table 4 as influencing this misalignment varies along those same dimensions. This is particularly relevant for understanding which specific policies are most needed in different areas of the EU, what types of policy interventions can be most reasonably expected to enhance the performance of exporters within rather than outside the single market, and how such measures are likely to interact with ongoing trends in international markets. This exercise is also essential for monitoring the extent to which EU member states have been converging over time, and, accordingly, for assessing the degree of policy differentiation that may still be required among them.

¹⁷The results for RST in column 1 essentially reflect equation 8, which indicates that, for a given level of observed exports, RST is higher when factors that hinder export activity, such as relatively high wages and low firm selection, are larger.

¹⁸Less firm heterogeneity, as captured by a higher Pareto shape parameter $k(z)$, implies fewer firms with relatively low core-competency UIR, that is, fewer firms with relatively high productivity.

Table 5: Estimation of Components within and outside the EU

RST-BRCA	All (1)	In EU (2)	Ex EU (3)
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	0.135*** (0.015)	0.130*** (0.014)	0.144*** (0.015)
Firms Heterogeneity $k(z)$	-0.397*** (0.020)	-0.388*** (0.020)	-0.417*** (0.020)
Trade Freeness ρ	-0.594*** (0.039)	-0.554*** (0.039)	-0.623*** (0.039)
Relative Market Size $\frac{L}{L^*}$	-0.094*** (0.027)	-0.109*** (0.026)	-0.085*** (0.027)
R^2	0.94	0.95	0.95
Obs.	2,576	2,576	2,576

Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Notes: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column 1 replicates column 3 in Table 4. A unit of observation is a year-country-industry combination. *Relative State of Technology (RST)* is computed like in equation (8). *Balassa Revealed CA (BRCA)* is computed like $[Exp^{c \rightarrow d}(z) / \sum_z Exp^{c \rightarrow d}(z)] / [Exp^{World}(z) / \sum_z Exp^{World}(z)]$ where $Exp^{c \rightarrow d}(z)$ are total exports in industry z from country c to destination d , and d is RoW in column 1, the rest of the EU in column 2, and RoW outside of the EU in column 3. For each measure, the value corresponds to the ranking of the industry within a given country and year, and *RST-BRCA* is the difference between the respective rankings. All regressions include country-industry and year fixed effects. Estimations for HR, CZ, DK, FI, FR, DE, HU, LT, MT, NL, PL, PT, RO, SK, SI, and SE between 2010 and 2018.

Table 6: Estimation of Components within and outside the EU through Time

RST-BRCA	All Countries			CESEE		
	All (1)	In EU (2)	Ex EU (3)	All (4)	In EU (5)	Ex EU (6)
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	0.151*** (0.018)	0.139*** (0.018)	0.170*** (0.018)	0.058*** (0.022)	0.052** (0.021)	0.079*** (0.022)
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)} \times 2013 - 2015$	-0.003 (0.018)	0.001 (0.018)	-0.020 (0.018)	-0.008 (0.025)	-0.003 (0.024)	-0.015 (0.024)
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)} \times 2016 - 2018$	-0.045** (0.019)	-0.024 (0.018)	-0.053*** (0.019)	-0.053** (0.026)	-0.022 (0.026)	-0.056** (0.026)
Firms Heterogeneity $k(z)$	-0.346*** (0.023)	-0.342*** (0.023)	-0.341*** (0.023)	-0.257*** (0.031)	-0.273*** (0.031)	-0.265*** (0.031)
Firms Heterogeneity $k(z) \times 2013 - 2015$	-0.090*** (0.019)	-0.075*** (0.019)	-0.129*** (0.019)	-0.140*** (0.026)	-0.118*** (0.025)	-0.162*** (0.025)
Firms Heterogeneity $k(z) \times 2016 - 2018$	-0.070*** (0.019)	-0.073*** (0.019)	-0.090*** (0.019)	-0.126*** (0.026)	-0.117*** (0.026)	-0.128*** (0.026)
Trade Freeness ρ	-0.607*** (0.041)	-0.561*** (0.041)	-0.639*** (0.041)	-0.719*** (0.049)	-0.671*** (0.049)	-0.719*** (0.049)
Trade Freeness $\rho \times$ 2013-2015	0.074*** (0.025)	0.074*** (0.025)	0.097*** (0.025)	0.103*** (0.035)	0.113*** (0.034)	0.122*** (0.034)
Trade Freeness $\rho \times$ 2016-2018	0.037 (0.026)	0.019 (0.026)	0.035 (0.026)	0.069* (0.037)	0.050 (0.036)	0.050 (0.036)
Relative Market Size $\frac{L}{L^*}$	-0.123*** (0.030)	-0.127*** (0.030)	-0.137*** (0.030)	-0.126*** (0.039)	-0.136*** (0.038)	-0.138*** (0.038)
Relative Market Size $\frac{L}{L^*} \times 2013 - 2015$	0.027 (0.025)	0.018 (0.025)	0.057** (0.025)	0.026 (0.034)	0.011 (0.034)	0.066* (0.034)
Relative Market Size $\frac{L}{L^*} \times 2016 - 2018$	0.021 (0.026)	0.002 (0.025)	0.045* (0.025)	0.026 (0.035)	0.009 (0.035)	0.042 (0.035)
R^2	0.95	0.95	0.95	0.95	0.95	0.95
Obs.	2,576	2,576	2,576	1,363	1,363	1,363

Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Notes: Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Estimations for all countries in our sample (columns 1-3) and CESEE countries only (columns 4-6). A unit of observation is a year-country-industry combination. *Relative State of Technology (RST)* is computed like in equation (8). *Balassa Revealed CA (BRCA)* is computed like $[Exp^{c \rightarrow d}(z) / \sum_z Exp^{c \rightarrow d}(z)] / [Exp^{World}(z) / \sum_z Exp^{World}(z)]$ where $Exp^{c \rightarrow d}(z)$ are total exports in industry z from country c to destination d , and d is RoW in columns 1 and 4, the rest of the EU in columns 2 and 5, and RoW outside of the EU in columns 3 and 6. Baseline for 2010-2012. *2013-2015* and *2016-2018* are dummies for the respective years. For each measure, the value corresponds to the ranking of the industry within a given country and year, and *RST-BRCA* is the difference between the respective rankings. CESEE countries are HR, CZ, HU, LT, PL, RO, SK, and SI. Non-CESEE countries are DK, FI, FR, DE, MT, NL, PT, and SE.

Table 5 disentangles the findings in Table 4 by destination. Compared with the general results, we observe that relative market size is more relevant in driving the misalignment between RST and BRCA within the single market, whereas relative unit input prices, firm heterogeneity, and trade freeness play a greater role outside the EU. When further disaggregating by destination, CESEE country status, and time in Table 6, we find that the magnitude of trade freeness has generally been declining for both internal and external destinations, while that of relative unit input prices and relative market size has decreased significantly only vis-à-vis extra-EU markets. By contrast, firm heterogeneity has exhibited an upward tendency across the board. For the CESEE sub-sample, patterns are similar, and accentuated time trends have closed initial gaps so that, by the end of the period (2018), it had converged with the rest of the EU in terms of the roles of firm heterogeneity, trade freeness, and relative market size. However, the fact that relative unit input prices have lost strength in the comparison between extra-EU export specialization and RST, though not in the corresponding within-EU comparison, has almost completely abated the role of relative factor prices for CESEE countries, but not within the single market.

5 Conclusion

Numerous reports analyzing the economic performance of the European Union (most recently the Draghi report [Draghi, 2024](#)) have put technological prowess at the center of the policy debate on EU competitiveness. Based on the sufficient statistics approach developed by [Huang and Ottaviano \(2024\)](#), we have shown how the state of technology of European industries relative to the rest of the world can be empirically assessed in a way that is simple in terms of computation, parsimonious in terms of data requirements, but still comprehensive in terms of information. This has allowed us to answer two main questions on where Europe’s relative technological advantage currently lies, and whether market forces may misalign its relative technological advantage and its relative export specialization.

We have found that Europe has better state of technology than the rest of the world in all manufacturing industries, but especially in wearing apparel, food, basic metals and furniture, and non-metallic mineral products. Among these industries, however, Europe also exhibits export specialization only in food and non-metallic mineral products. In other industries, such as motor vehicles, wood, paper products, and machinery, Europe has strong export specialization despite less pronounced technological advantage.

In general, the lack of systematic cross-industry correlation between relative export specialization and technological advantage suggests that standard measures of revealed comparative advantage only imperfectly capture a country’s technological prowess due to the concurrent influences of factor prices, market size, markups, firm selection and market share reallocation. This can be explained in terms of competitive pressures and scale effects associated with national market size and international trade.

These findings offer policy insights relevant to the EU’s external competitiveness debate, echoing several recommendations from the Draghi report. Achieving export specialization in key sectors requires more than just technological superiority. It also necessitates favorable conditions to capitalize on that advantage, including affordable access to necessary production factors and intermediate inputs, a substantial domestic market for related goods, and a dynamic industrial environment that facilitates the introduction of these products. Therefore, policy efforts should strategically target the development of workforce skills, the effective deployment of investment capital, the intelligent structuring of supply chains, the continued deepening of single market integration by encouraging labor and capital flows, and the simplification of firm entry and exit processes.

We find that these considerations have become applicable to different EU countries with increasingly similar expected outcomes as a result of convergence. Moreover, we observe that they are particularly relevant for global markets beyond the EU, where the disconnect between technological and export performance is more pronounced, has increased over time, and is characterized by a stronger influence of relative factor prices, trade freeness, and the emergence of consolidated firms within the population structure. The trend toward a growing prominence of the latter, in contrast to the lessening relevance of the former two, suggests that in the current conjuncture, the increasingly challenging task of maintaining access to extra-EU markets may be

partially undermined if European industries are not adequately positioned to enable the emergence of firms capable of seizing these opportunities.

By construction, the sufficient statistics approach adopted in this paper is well suited for the ex-post evaluation of the effects of implemented policies, but has little to say about the ex-ante assessment of the possible effects of proposed policies yet to be implemented. Future research should then focus on developing a consistent framework for ex-ante assessment. This framework should go beyond the two-country setup, and may lead to the identification of a set of sufficient statistics to be used in a full-fledged multi-country setup, thus making the two-country aggregation of this paper unnecessary.

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Appendices

A CompNet Data Coverage

Table A.1: CompNet Coverage against Eurostat SBS and OECD ICIO, by Key Variables

	Employment	Revenues	Export	Labor Productivity
Country				
Croatia	0.82	0.93	1.35	1.15
Czech Republic	0.97	0.92	0.82	1.01
Denmark	0.85	0.90	1.24	1.11
Finland	0.87	0.80	0.77	0.96
France	0.75	0.69	0.89	0.91
Germany	0.96	1.03	1.07	1.03
Hungary	0.88	0.84	1.03	0.95
Lithuania	0.84	0.88	0.85	1.04
Malta	1.77	3.54	0.82	0.62
Netherlands	0.87	0.89	0.80	0.89
Poland	0.88	0.86	0.88	0.98
Portugal	0.74	0.73	0.84	0.92
Romania	0.87	0.82	1.32	0.99
Slovakia	0.88	0.79	0.78	0.95
Slovenia	0.84	0.88	0.94	1.06
Sweden	0.90	0.92	1.05	1.13
Industry				
10T12-Food products, beverages and tobacco	0.82	0.91	1.02	1.05
13T15-Textiles, textile products, leather and footwear	0.92	1.07	1.46	1.17
16-Wood and products of wood and cork	0.70	0.79	0.83	1.26
17_18-Paper products and printing	0.83	0.90	0.91	1.05
20-Chemical and chemical products	1.00	0.93	0.95	0.90
21-Pharmaceuticals, medicinal chemical and botanical products	1.08	0.83	1.18	1.04
22-Rubber and plastics products	0.95	1.01	0.95	1.06
23-Other non-metallic mineral products	0.84	0.87	0.91	1.12
24-Basic metals	0.95	0.85	0.76	0.94
25-Fabricated metal products	0.79	0.85	1.00	1.16
26-Computer, electronic and optical equipment	0.95	0.77	0.80	0.68
27-Electrical equipment	1.00	1.07	0.97	1.05
28-Machinery and equipment, nec	0.99	1.05	1.01	1.05
29-Motor vehicles, trailers and semi-trailers	0.94	0.87	1.06	0.92
30-Other transport equipment	0.95	0.84	1.02	1.02
31T33-Manufacturing nec; repair and installation of machinery and equipment	0.76	0.81	1.01	1.18
Year				
2010	0.87	0.90	0.91	1.01
2011	0.88	0.92	0.86	1.05
2012	0.90	0.91	0.95	1.01
2013	0.90	0.92	0.92	1.03
2014	0.90	0.90	0.89	0.98
2015	0.91	0.93	1.10	1.01
2016	0.89	0.89	1.09	0.96
2017	0.89	0.89	1.08	0.97
2018	0.84	0.89	1.04	0.99

Source: CompNet, Eurostat SBS, OECD ICIO.

Notes: *Employment*, *Revenues*, and *Labor Productivity* are the ratios of the respective measures in CompNet over Eurostat SBS for the matched year-country-industry combinations. *Labor Productivity* is computed like revenue per employee. *Export* is the ratio of the respective measure in CompNet over OECD ICIO for the matched year-country-industry combinations. The panel *Country* reports the ratio for the sums of *Employment*, *Revenues*, and *Export* and for the means of *Labor Productivity* by country over industries and years. The panel *Industry* reports the ratio for the sums of *Employment*, *Revenues*, and *Export* and for the means of *Labor Productivity* by industry over countries and years. The considered year-country-industry combinations are for the countries and industries in the table, and for years between 2010 and 2018.

B Summary Statistics

Table B.1: Summary Statistics for Key Variables

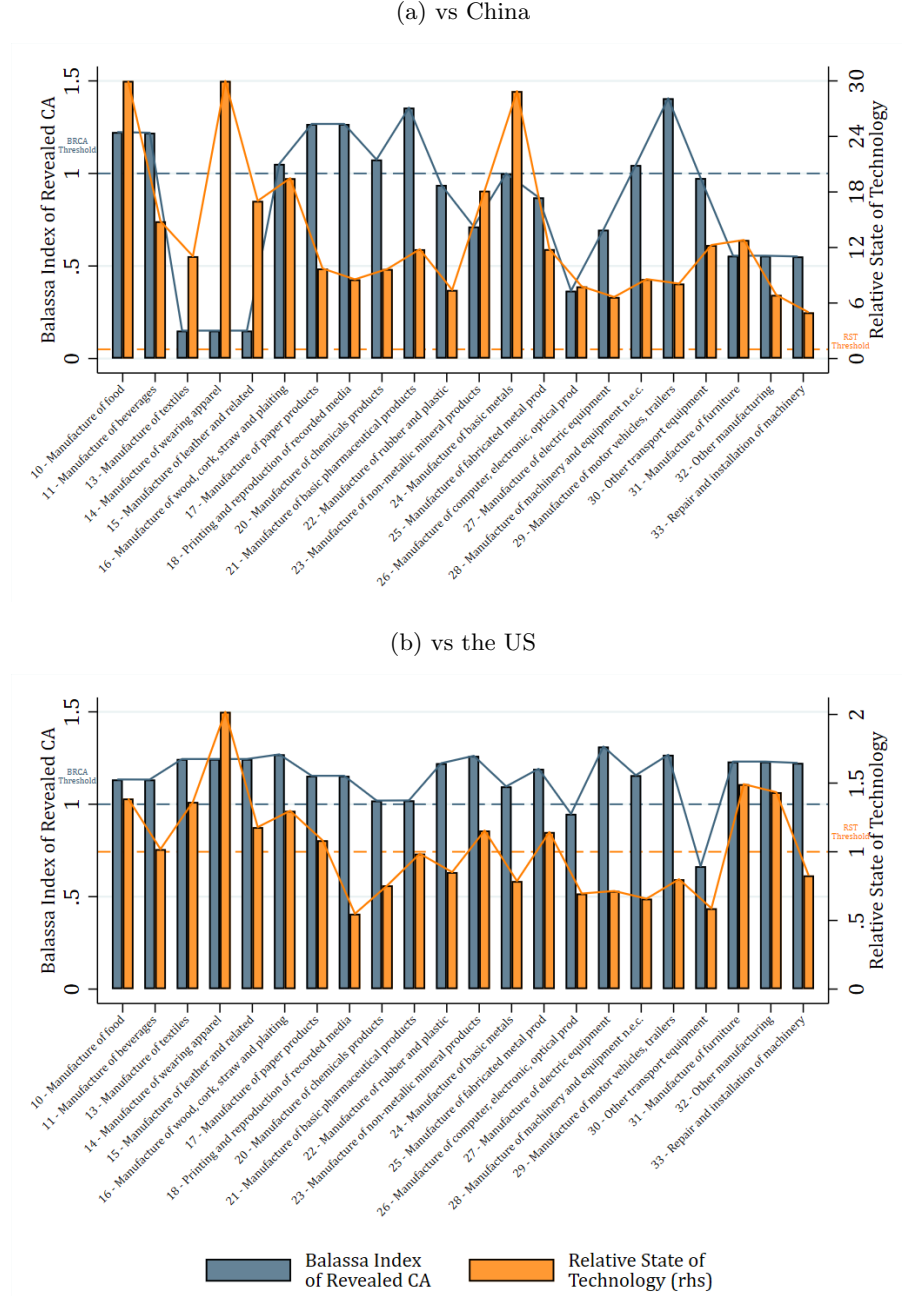
Variable	Obs.	Mean	P10	P25	P50	P75	P90	SD
All Countries								
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	2,578	1.54	1.16	1.29	1.42	1.75	2.10	0.35
Trade Freeness ρ	2,578	0.43	0.17	0.25	0.40	0.57	0.74	0.23
Firms Heterogeneity $k(z)$	2,578	1.15	0.75	0.93	1.12	1.39	1.59	0.32
Relative Market Size $\frac{L}{L^*}$	2,578	4.93	0.28	0.63	1.39	3.98	11.21	14.47
European Union								
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	193	1.95	1.69	1.82	1.92	2.03	2.24	0.21
Trade Freeness ρ	193	0.39	0.17	0.23	0.33	0.54	0.63	0.19
Firms Heterogeneity $k(z)$	193	1.15	0.70	0.93	1.12	1.38	1.54	0.31
Relative Market Size $\frac{L}{L^*}$	193	3.29	0.59	0.93	2.25	4.12	8.31	3.34
France								
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	196	1.27	1.12	1.19	1.27	1.35	1.43	0.13
Trade Freeness ρ	196	0.38	0.16	0.19	0.28	0.56	0.70	0.25
Firms Heterogeneity $k(z)$	196	1.16	0.74	0.93	1.13	1.39	1.62	0.32
Relative Market Size $\frac{L}{L^*}$	196	4.60	0.61	1.11	2.68	6.90	11.21	4.45
Germany								
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	198	2.01	1.75	1.97	2.04	2.12	2.18	0.17
Trade Freeness ρ	198	0.37	0.16	0.23	0.29	0.53	0.65	0.19
Firms Heterogeneity $k(z)$	198	1.16	0.74	0.93	1.13	1.39	1.62	0.32
Relative Market Size $\frac{L}{L^*}$	198	4.48	0.80	1.01	2.93	6.57	10.00	4.51
Poland								
Relative Unit Input Price $\frac{\omega(z)}{\omega^*(z)}$	187	1.73	1.70	1.70	1.72	1.75	1.79	0.04
Trade Freeness ρ	187	0.45	0.19	0.26	0.41	0.62	0.75	0.22
Firms Heterogeneity $k(z)$	187	1.15	0.73	0.92	1.11	1.40	1.63	0.33
Relative Market Size $\frac{L}{L^*}$	187	2.96	0.22	0.85	1.52	2.50	4.63	6.06

Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Notes: *Relative Unit Input Price*, *Trade Freeness*, *Firms Heterogeneity*, and *Relative Market Size* are computed like in subsection 4.1. The unit of observation is a year-country-industry combination, for counties, industries, and years in Table A.1, the only exception being *Firms Heterogeneity* which is computed like the year-industry average. The figures for the *European Union* are derived by aggregating variables across the EU countries included in our sample: HR, CZ, DK, FI, FR, DE, HU, LT, MT, NL, PL, PT, RO, SK, SI, and SE.

C Comparisons with China and the US

Figure C.1: Balassa Revealed Comparative Advantage and Relative State of Technology in the EU



Source: CompNet, OECD ICIO, Penn World Table, and calculations by the authors.

Note: *Relative State of Technology* is computed like in equation (8), but restricting the definition of RoW to China (panel a) or the US (panel b). *Balassa Index of Revealed CA* is analogously computed by restricting the comparison of export specialization to China (panel a) or the US (panel b), i.e., with either $P = \text{China}$ or $P = \text{US}$, like $[Exp^{EU}(z)/\sum_z Exp^{EU}(z)]/[Exp^{EU+P}(z)/\sum_z Exp^{EU+P}(z)]$ where $Exp^X(z)$ are total exports in industry z from X to RoW. *Balassa Index of Revealed CA* and *Relative State of Technology* are represented on the left-hand and right-hand axes, respectively. The dashed blue line represents the threshold for the *Balassa Index of Revealed CA* to indicate specialization. The dashed orange line represents the threshold for the *Relative State of Technology* to indicate technological advantage. The figures for the EU are derived by aggregating components (exports, export intensity, export propensity, trade freeness, and relative unit input prices) across the EU countries included in our sample: HR, CZ, DK, FI, FR, DE, HU, LT, MT, NL, PL, PT, RO, SK, SI, and SE. The OECD ICIO tables consolidate sectors 10, 11, 13, 14, 15, 31, 32, and 33 into broader categories. For these sectors, the *Balassa Index of Revealed CA* for the corresponding category is reported. Average from 2010 to 2018.